Outside programs and AI agents used: No AI agents were used to write words on this document. For questions involved with demand estimation and counterfactuals, we first attempt to solve using "manual" coding using online guides (footnotes indicate where online guides were followed closely) and then re-estimated using PyBLP. This allows us to understand the mechanics behind the models we study and how to use standardized tools that are increasingly used in empirical industrial organization.

Problem 1

We estimate the parameters in the utility function for each model in turn. Further, we include certain derivations beyond the strict scope of the questions for our pedagogical understanding (e.g., deriving mean utility as the difference in log market shares for logit demand)[1]:

- Logit demand:

[1]For the PyBLP implementation, we follow PyBLP's documentation (link here) and demand estimation exercises led by Jeff Gortmaker (link here)